Assignment 5: DAO Smart Contract

Project Overview

This project implements a Decentralized Autonomous Organization (DAO) smart contract. It allows users to create proposals, vote on them, and execute decisions in a decentralized manner.

Objective

To understand DAO working, voting mechanisms, and proposal lifecycle using blockchain smart contracts.

DAO Smart Contract

The DAO is implemented using Solidity. It includes functions for proposal creation, voting, and execution.

Voting Mechanism

Each member can vote on proposals. Votes are recorded on-chain and counted to determine the outcome. Majority voting is used for decision making.

Proposal Creation

Users can create proposals by submitting details. Each proposal has a voting period during which members can vote.

Workflow of DAO

1. Deploy smart contract
2. Create proposal
3. Members vote
4. Votes counted
5. Proposal executed

Screenshots

```
PS C:\Users\Sohan Metil\Desktop\bt3> npx hardhat run assignment-5/deploy.js --network localhost
Deploying SimpleDAO...
✅ DAO deployed to: 0xe7f1725E7734CE288F8367e1Bb143E90bb3F0512
PS C:\Users\Sohan Metil\Desktop\bt3>
```

Conclusion

This assignment demonstrates how DAOs work using blockchain technology with transparent and decentralized governance.